WHAT IF?! EP 02

FREE GUIDE
you commented FLUSH

The Flush Fact Sheet



Plume rules, the never-flush list and a quick plumbing quiz.
Everything the reel waved at, with the numbers and where they came from.

PLUME RULES

1 Know the splash zone

A flush launches droplets up to about **0.9 m (3 ft)** high and they can hang around for over a minute (2020 model). Your toothbrush lives in that zone.

2 Lid down, always

In a lidless-toilet experiment, droplets peaked about a foot above the bowl within roughly 5.5 seconds of the flush. Close the lid first.

3 A lid is not a seal

A later pilot study found aerosols still escape through the seat gap with the lid down: 24% to 57% depending on the microbe, at a 6 mm gap.

4 Do not panic

A 2013 review says plumes *could* contribute to spreading infection; more research is needed. Rules 1 to 3 are the cheap version.

THE NEVER-FLUSH LIST

X Wet wipes

The main ingredient of London's 2017 Whitechapel fatberg.



X Fat, oil and grease

The glue that sets the wipes into a lump.



Those two built the fatberg: about **130 tonnes**, over **250 m long** (more than two football pitches), mostly wet wipes plus fat. Part of it went on display at the Museum of London.

THE NUMBERS

~30%
of indoor home water goes down the toilet (US EPA WaterSense)

4.8 L
a modern 1.28 gal flush. Pre-1980: about 19 L

3.4 bn
people still lack safely managed sanitation (2024, WHO/UNICEF)

1983
M*A*S*H finale: reportedly ~6.7 million gallons extra into New York sewers

+6,477
gallons a minute in Seattle in the 15 min after the 2026 Super Bowl halftime show

1858
London's Great Stink: Parliament soaked its curtains in chloride of lime

Sources Physics of Fluids 2020 and Crimaldi et al., Sci Reports (plume); Johnson et al., Am J Infect Control 2013 · London Museum: Whitechapel fatberg, Great Stink · US EPA WaterSense · WHO/UNICEF JMP (data.unicef.org) · UPI 14 Mar 1983 · NYC DEP Super Flush; KIRO 7

Quick plumbing quiz

Tick one answer per question, then check the key and give yourself a point for each hit.



1. **How high can a flush plume rise from an open toilet?**
   - ☐ A. about 10 cm
   - ☐ B. about 0.9 m
   - ☐ C. about 3 m
2. **After the 1983 M*A*S*H finale, New York reportedly added about how much to its sewers?**
   - ☐ A. 67 gallons
   - ☐ B. 67,000 gallons
   - ☐ C. 6.7 million gallons
3. **How heavy was London's 2017 Whitechapel fatberg?**
   - ☐ A. about 13 tonnes
   - ☐ B. about 130 tonnes
   - ☐ C. about 1,300 tonnes
4. **What share of a home's indoor water goes down the toilet?**
   - ☐ A. about 3%
   - ☐ B. about 30%
   - ☐ C. about 60%
5. **In Australia, which way does a toilet flush?**
   - ☐ A. backwards
   - ☐ B. depends on the bowl design
   - ☐ C. straight up
6. **How many people still lack safely managed sanitation (2024)?**
   - ☐ A. 3.4 million
   - ☐ B. 340 million
   - ☐ C. 3.4 billion
7. **A modern WaterSense toilet uses at most about how much per flush?**
   - ☐ A. 4.8 litres
   - ☐ B. 13 litres
   - ☐ C. 19 litres
8. **Has a Super Bowl flush ever broken a sewer system?**
   - ☐ A. yes, most years
   - ☐ B. no: utilities just prepare for the demand
   - ☐ C. only in Seattle

ANSWER KEY

1 B. About 0.9 m (3 ft), per the 2020 model.

2 C. About 6.7 million gallons, as reported by UPI.

3 B. About 130 tonnes, over 250 m long.

4 B. Close to 30% (US EPA WaterSense).

5 B. Swirl comes from bowl design and rim jets, not Australia.

6 C. About 3.4 billion, per WHO/UNICEF.

7 A. 1.28 gal (about 4.8 L) or less.

8 B. NYC pre-sends water, but no sewer has failed from it.

YOUR SCORE

0-2	3-5	6-7	8
Wet-wipe wanderer	Apprentice plumber	Sewer sage	Fatberg slayer



Three things we did not claim

"Billions of toilets": nobody has counted flush toilets. "Sewers would explode": even the Super Bowl surge never has. "The 1984 Salt Lake City main break proves it": no link was ever found. The what-if is a cartoon; the plumbing facts are real.

Sources UPI 14 Mar 1983 · London Museum (Whitechapel fatberg) · US EPA WaterSense (residential toilets) · Live Science / Mental Floss (flush direction) · WHO/UNICEF JMP 2024 · NYC DEP news 25-006; WUSA9 Verify